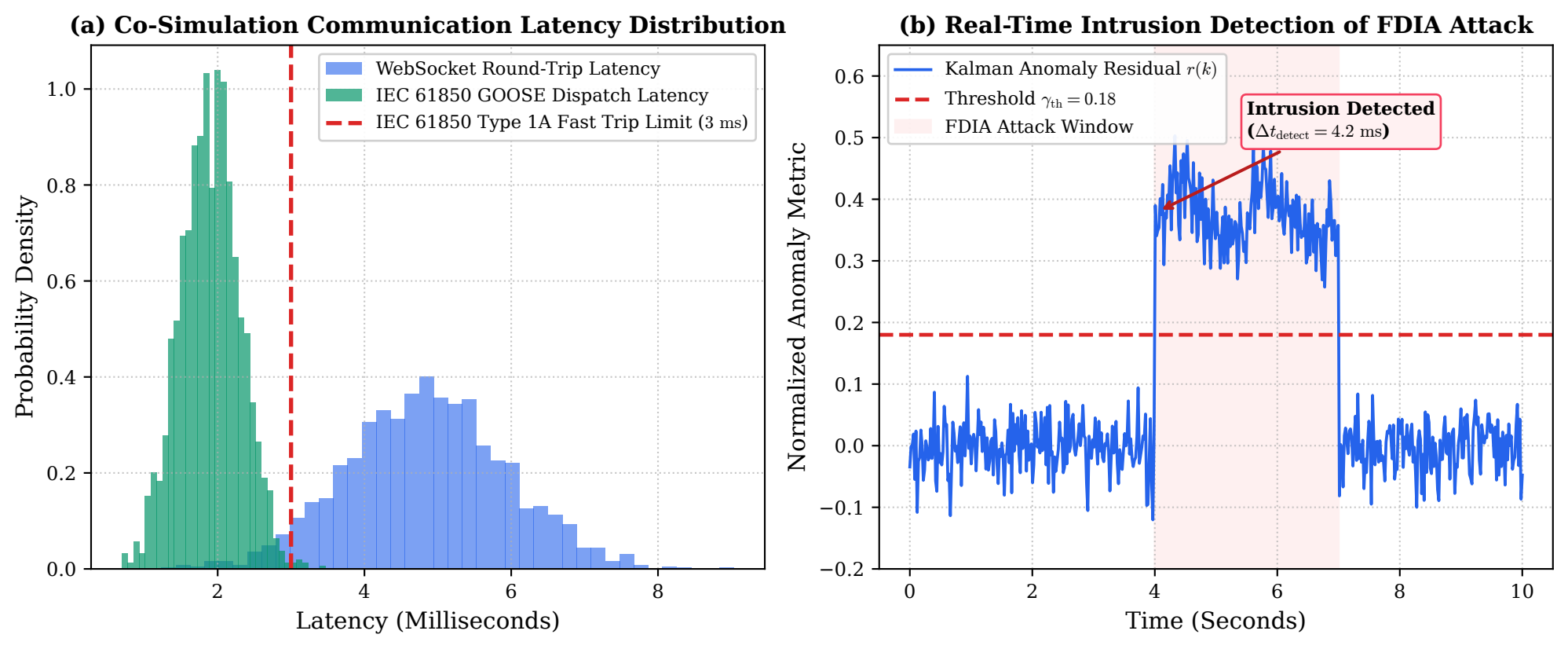

(a) Co-Simulation Communication Latency Distribution
WebSocket Round-Trip Latency
IEC 61850 GOOSE Dispatch Latency
IEC 61850 Type 1A Fast Trip Limit (3 ms)
Probability Density
Latency (Milliseconds)
(b) Real-Time Intrusion Detection of FDIA Attack
Kalman Anomaly Residual $r(k)$
Threshold $\gamma_{\text{th}} = 0.18$
FDIA Attack Window
Intrusion Detected
($\Delta t_{\text{detect}} = 4.2$ ms)
Normalized Anomaly Metric
Time (Seconds)